Discrete Mathematics and Statistics (CPT107)
Tutorial 2 - Solutions

1: If X and Y are empty sets, then X = Y. If you think that it is true, prove it. If not, explain why.

Solution: This is true. Assume that X and Y are not equal, then we should be able to find an element from set X which does not belong to Y, but X contains no element. Similarly, we cannot find an element in Y that does not belong to X (again Y contains no elements at all).

2: Let X, Y and Z be sets. If $X \subset Y$ and $Y \subset Z$, then $X \subset Z$. If you think that it is true, prove it. If not, explain why.

Solution: This is true. Let $x \in X$. Since $X \subset Y$ and $x \in X$, we can conclude that $x \in Y$. Since $Y \subset Z$ and $x \in Y$, we can conclude that $x \in Z$ and therefore $X \subset Z$.

3: Let X, Y and Z be sets. Is this expression $(X-Y)\cup(Y-Z)=X-Z$ true? If your answer is yes, then prove it. Otherwise, give a counterexample.

Solution:

No. A counterexample can be $X=\{1,2,4,5\}$, $Y=\{2,3,5,6\}$, $Z=\{4,5,6,7\}$

$X-Y=\{1,4\}$;

$Y-Z=\{2,3\}$;

$(X-Y)\cup(Y-Z)=\{1,2,3,4\}$;

$X-Z=\{1,2\}$

4: Prove that the following sets are equal:

1. $A \times (B \cup C) = (A \times B) \cup (A \times C)$
2. $A \times (B \cap C) = (A \times B) \cap (A \times C)$
3. $A \times (B - C) = (A \times B) - (A \times C)$

Solution:

$$A \times (B \cup C) = (A \times B) \cup (A \times C)$$

- Let (a,x) ∈ A x (B ∪ C). This means that a ∈ A and x ∈ B ∪ C. It logically follows by the distributive laws that a ∈ A and x ∈ B, or a ∈ A and x ∈ C. This means that (a,x) ∈ (AxB) ∪ (AxC). Thus every element of A x (B ∪ C) is an element of (AxB) ∪ (AxC). Therefore, A x (B ∪ C) ⊆ (A x B) ∪ (A x C).
- For the reverse containment, let (a,x) ∈ (A x B) ∪ (A x C). This means that either a ∈ A and x ∈ B, or a ∈ A and x ∈ C. It follows that a ∈ A, and either x ∈ B or x ∈ C. Thus, (a,x) ∈ A x (B ∪ C). So (A x B) ∪ (A x C) ⊆ A x (B ∪ C). Since both containments hold it follows that the sets in question are equal.
- The proof of the other ones are similar.

5: Let A and B be two sets. Prove the following statements: $A \subseteq B$ if and only if $\sim B \subseteq \sim A$.

Solution:

<u>If $A \subseteq B$ then $\sim B \subseteq \sim A$:</u>

Assume $A \subseteq B$. This means that if $x \in A$, then $x \in B$. Let $y \in \sim B$. This means that $y \notin B$, Then, it follows that $y \notin A$, which means that $y \in \sim A$. Hence, we have shown $\sim B \subseteq \sim A$.

<u>If $\sim B \subseteq \sim A$ then $A \subseteq B$:</u>

Apply the forward direction with $\sim B$ and $\sim A$ in place of A and B, respectively, and use the fact that $\sim (\sim C) = C$.